

Computer Vision – Individual Lab 1

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Problem

In this assignment, you only use 3 OpenCV functions: *imread*, *imwrite*, and *imshow*. You have to implement from scratch several image manipulations using OpenCV C/C++:

- Convert a color image to the gray image
- Change the brightness/contrast of an image
- Filter an image
- Edge detection

Your program is called by command line arguments and perform the following functions

1. Load the input image from its path
2. Show the result image and save it
3. Convert a color image to the gray image. The command line is

<Executable file> -rgb2gray <InputFilePath> <OutputFilePath>

where

- *-rgb2gray*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file

4. Change the brightness of an image. The command line is

<Executable file> -brightness <InputFilePath> <OutputFilePath> <C>

where

- *-brightness*: the name of command

- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file
- *C*: the brightness factor

5. Change the contrast of an image. The command line is

```
<Executable file> -contrast <InputFilePath> <OutputFilePath> <C>
```

where

- *-contrast*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file
- *C*: the contrast factor

6. Filter an image using average filter. The command line is

```
<Executable file> -avg <InputFilePath> <OutputFilePath> <k>
```

where

- *-avg*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file
- *K*: the size of kernel filter

7. Filter an image using median filter. The command line is

```
<Executable file> -med <InputFilePath> <OutputFilePath> <k>
```

where

- *-med*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file
- *K*: the size of kernel filter

8. Filter an image using gaussian filter. The command line is

```
<Executable file> -gau <InputFilePath> <OutputFilePath> <k>
```

where

- *-gau*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file
- *K*: the size of kernel filter

9. Detect edge of an image using Sobel of kernel size 3×3 . The command line is

```
<Executable file > -sobel <InputFilePath> <OutputFilePath>
```

where

- *-sobel*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file

10. Detect edge of an image using Laplace of kernel size 3×3 . The command line is

```
<Executable file > -laplace <InputFilePath> <OutputFilePath>
```

where

- *-laplace*: the name of command
- *InputFilePath*: the path of input file
- *OutputFilePath*: the path of output file

Submission

Your submission must organize into 2 folders and compressed in 1 file *StudentID.zip*

1. Document: contains your own report and user guide
2. Data: contains input images and output images)
3. Sources: contain source code (.hpp, .cpp) which can re-build successfully.
4. Executable (contain the executable file is named *StudentID*)